

PHOENIX – Threat Analysis Sequence Diagram

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Project: PHOENIX

Deliverable: Threat Analysis Integration – Sequence Diagram

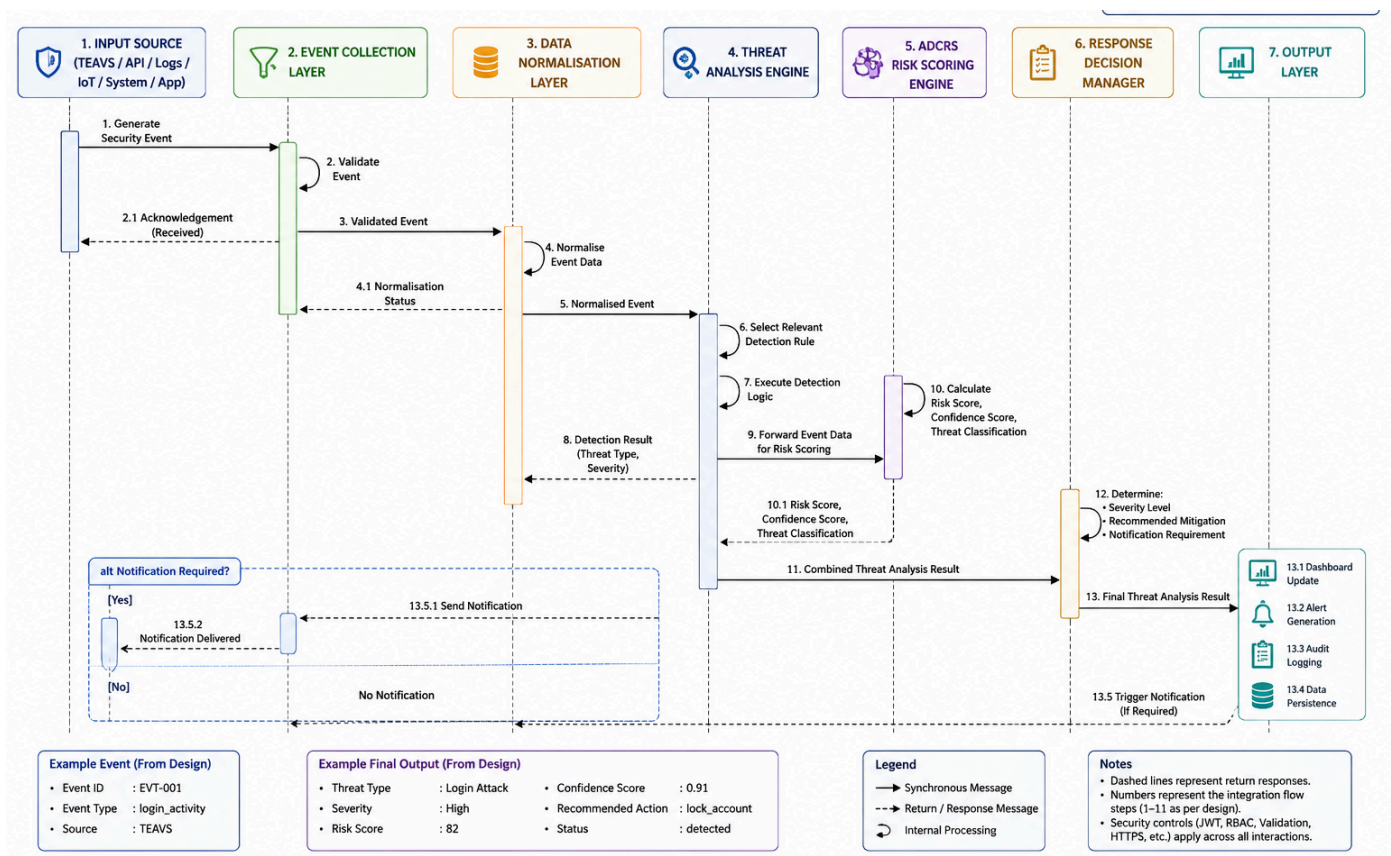
1. Purpose

The purpose of this sequence diagram is to illustrate the chronological interaction between the major components involved in the PHOENIX Threat Analysis Integration.

The diagram represents how a security event moves from an input source through event collection, data normalisation, rule-based threat analysis, ADCRS risk scoring and response decision processing before the final threat-analysis result is delivered to the output layer.

This sequence diagram was developed based on the approved Threat Analysis Integration Design Document and complements the PHOENIX Threat Analysis Integration Workflow Diagram.

2. Sequence Diagram



3. Sequence Description

Step 1 — Security Event Generation

A security event originates from one of the supported PHOENIX input sources, such as a TEAVS alert, API event, user activity log, IoT device event, system log or application event.

Step 2 — Event Collection and Validation

The Event Collection Layer receives the security event and validates the incoming event before it continues through the threat-analysis workflow.

Step 3 — Data Normalisation

The validated event is passed to the Data Normalisation Layer.

The event is converted into the common event structure containing:

- `event_id`
- `event_type`
- `timestamp`
- `source`
- `event_data`

Step 4 — Threat Analysis

The normalised event is provided to the Threat Analysis Engine.

The engine selects the relevant existing detection rule and executes the rule-based detection logic to identify the threat type and severity.

Step 5 — ADCRS Risk Scoring

Event data is forwarded to the ADCRS Risk Scoring Engine.

ADCRS provides:

- Risk Score
- Confidence Score
- Threat Classification

Step 6 — Combine Threat-Analysis Results

The results from the threat-detection process and ADCRS risk-scoring process are combined to produce the information required for response decision-making.

Step 7 — Response Decision

The Response Decision Manager processes the threat-analysis results and determines:

- Severity Level
- Recommended Mitigation
- Notification Requirements

Step 8 — Output Processing

The final threat-analysis result is passed to the Output Layer.

The Output Layer supports:

- Dashboard Updates
- Alert Generation
- Audit Logging
- Data Persistence

Notifications are triggered when required.

4. Security Considerations

Security controls defined for the Threat Analysis Integration apply throughout the sequence.

These include:

- JWT Authentication
- Role-Based Access Control (RBAC)
- Input Validation
- Rate Limiting
- Secure Error Handling
- Audit Logging
- Protected Routes
- HTTPS Communication

These controls help ensure that threat-analysis interactions are performed securely and that access, communication, event processing and auditing requirements are considered throughout the integration.

5. Example Sequence — Login Attack

An example event provided in the integration design can be used to demonstrate the sequence.

Incoming Event

EVT-001 → login_activity → TEAVS

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Event Collection

Receive and validate EVT-001

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Data Normalisation

Convert the event into the standard PHOENIX event structure.



Threat Analysis Engine

Apply the relevant rule-based detection logic.



ADCRS

Generate:

Risk Score: 82

Confidence Score: 0.91

Threat Classification: Login Attack



Response Decision Manager

Determine:

Severity: High

Recommended Action: lock_account



Output Layer

Produce the final result:

Status: detected

and support dashboard display, persistence, audit logging and notification where required.

6. Summary

The Threat Analysis Sequence Diagram provides a chronological representation of the interactions required to process a security event through the PHOENIX Threat Analysis Integration.

It builds upon the Integration Workflow Diagram by showing how information is transferred between each major integration component over time.

The sequence provides a technical reference for understanding component interactions and will also support the development of the **Integration Testing Plan**, where the individual interactions and expected results can be converted into functional and security validation scenarios.